

DIFFRACTION FROM N NUMBER OF SLITS

(DIFFRACTION GRATING)

Lecture #10

Content to be covered

- ❑ N-slits diffraction
 - ❖ Expression for resultant intensity
 - ❖ Position of maxima (Central and Secondary)
 - ❖ Position of minima
 - ❖ Intensity at secondary maxima
 - ❖ Spread of central diffraction maximum
- ❑ Diffraction Grating
 - ❖ Dispersive power of diffraction grating

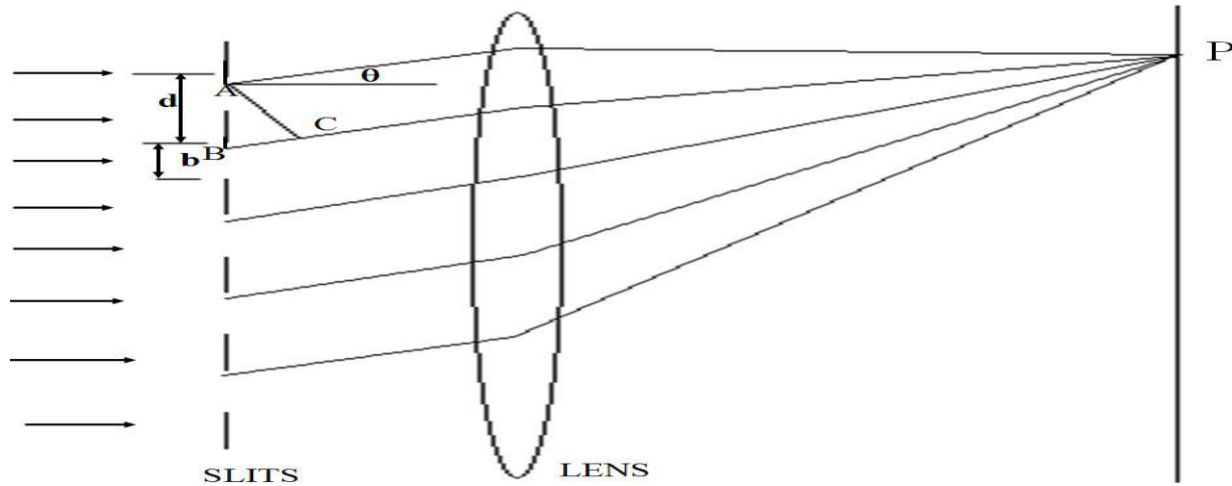
Resources to be consulted

- ❑ OPTICS by Ajay Ghatak

Diffraction from N number of slits (Diffraction Grating)

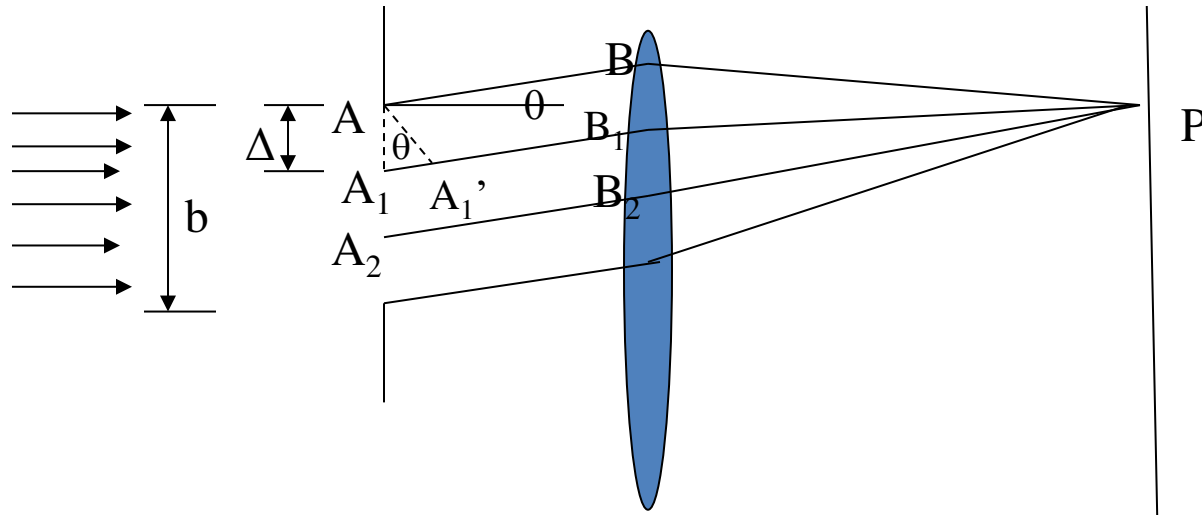
A plane diffraction grating is an arrangement **of a large number of close, parallel, straight, transparent and equidistant slits**, each of equal width ' b ', with neighboring slits being separated by an opaque region of width ' a '.

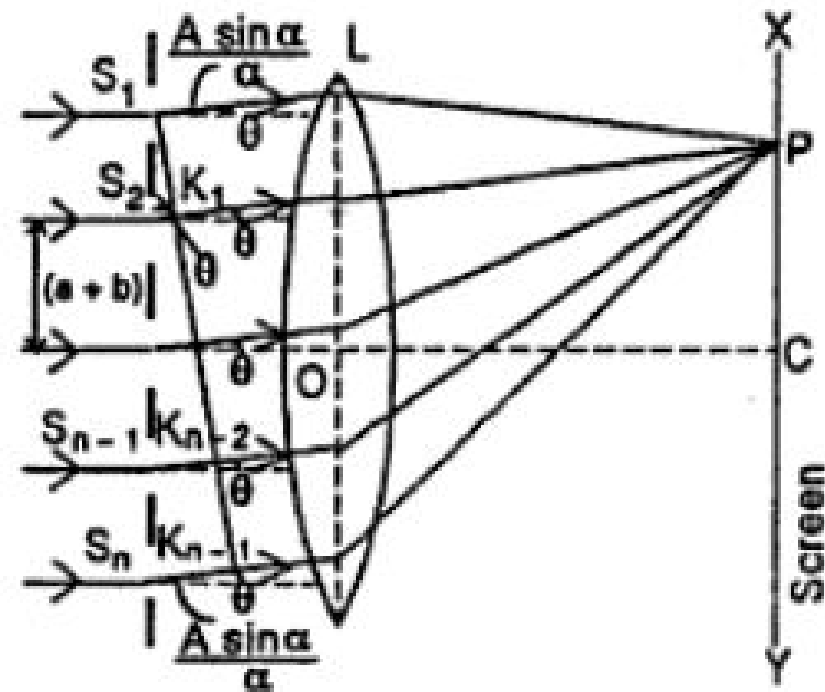
N slits



A plane diffraction grating is an arrangement **of a large number of close, parallel, straight, transparent and equidistant slits**, each of equal width ' b ', with neighboring slits being separated by an opaque region of width ' a '.

Single
slit

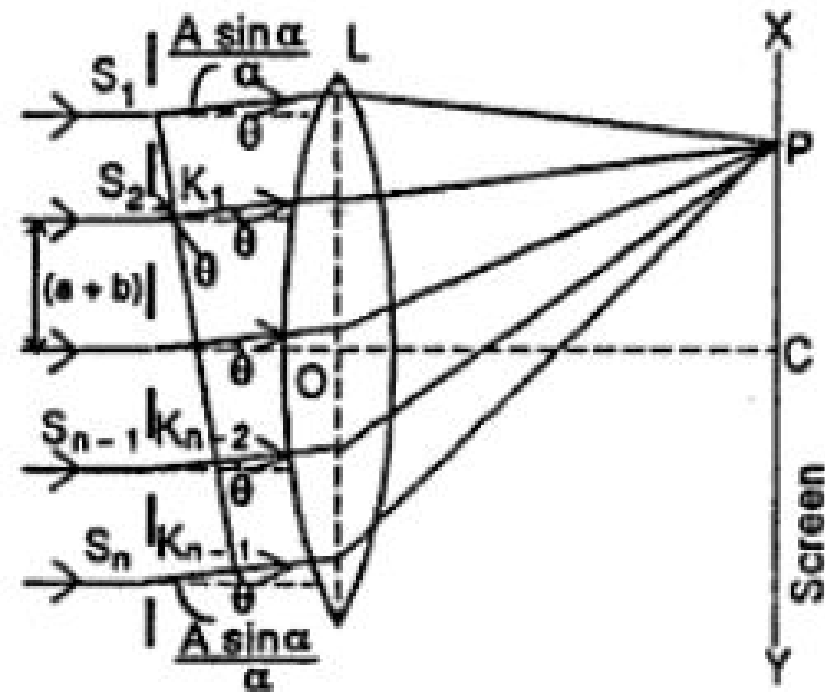




The resultant amplitude due to all waves diffracted from each slit on the direction θ is given by

$$R = \frac{A \sin \beta}{\beta}$$

where, $\beta = \frac{\pi b \sin \theta}{\lambda}$



The path difference between the waves originating from the slits S_1 and S_2 is

$$S_2K_1 = S_1S_2 \sin \theta = d \sin \theta$$

Similarly, between rays from S_2 and S_3 will be

$$S_3K_2 = S_2S_3 \sin \theta = d \sin \theta \text{ and so on...}$$

$$\Delta S_n K_{n-1} = S_{n-1}S_n \sin \theta = d \sin \theta$$

Hence, the resultant amplitude of 'N' waves of equal amplitude (R) and period but the phase ($\frac{2\pi d \sin \theta}{\lambda}$) increasing successively will be

$$R' = R \frac{\sin \frac{N\phi}{2}}{\sin \frac{\phi}{2}} = \frac{A \sin \beta}{\beta} \frac{\sin \frac{N\phi}{2}}{\sin \frac{\phi}{2}}$$

$$R = \frac{a \sin \frac{n\phi}{2}}{\sin \frac{\phi}{2}}$$

$$\Rightarrow I = \frac{A^2 \sin^2 \beta}{\beta^2} \frac{\sin^2 \frac{N\phi}{2}}{\sin^2 \frac{\phi}{2}}$$

where, $\gamma = \frac{\phi}{2} = \frac{\pi d \sin \theta}{\lambda}$

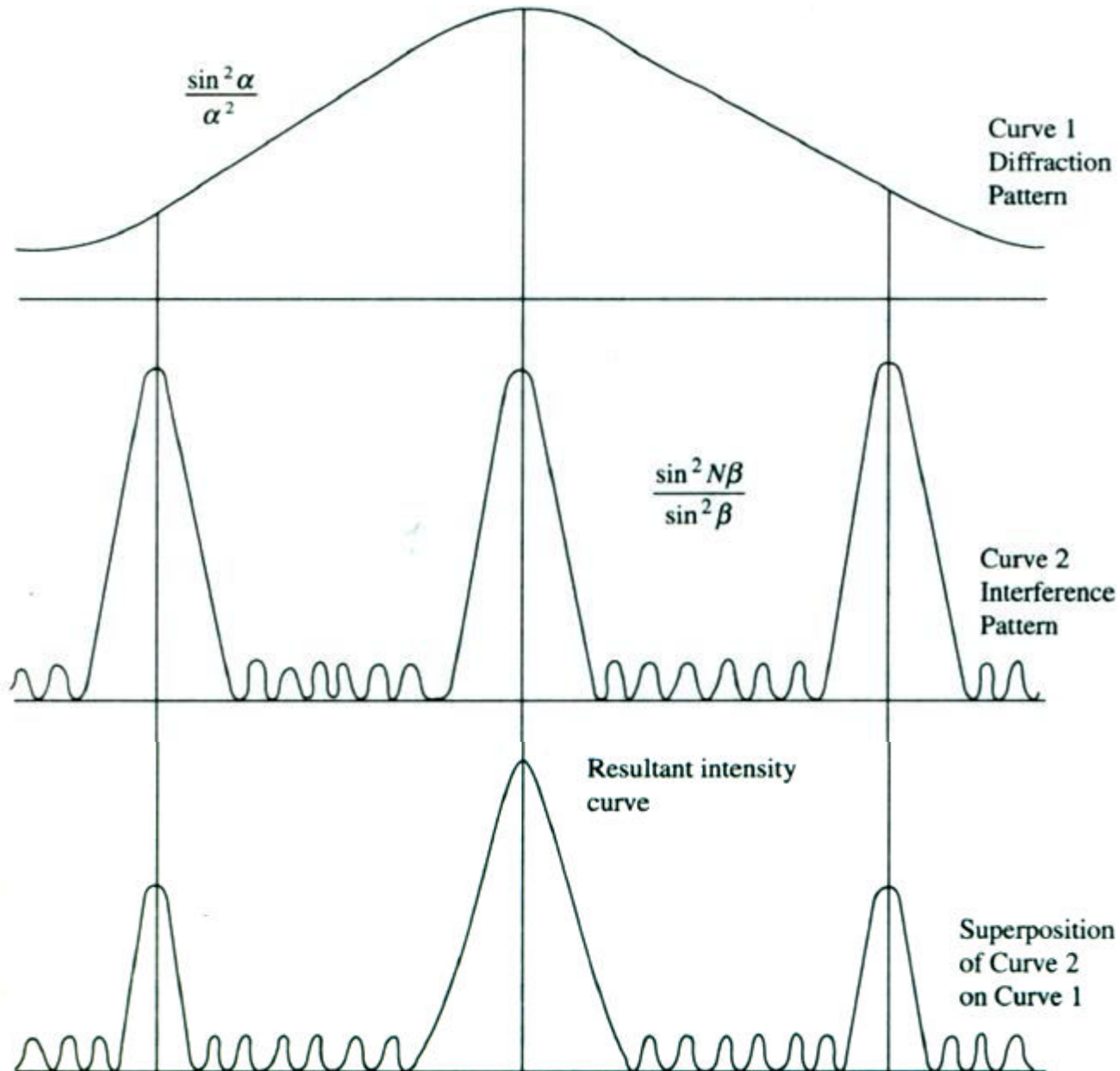
$\beta = \frac{\pi b \sin \theta}{\lambda}$

$I = \frac{A^2 \sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma}$

Represents the interference pattern produced by N equally spaced point sources.

represents the intensity distribution produced by a single slit.

Resultant intensity distribution of grating spectra



MAXIMA AND MINIMA

MAXIMA (Principal Maxima)

Intensity of diffraction pattern is given as

$$I = \frac{A^2 \sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma} \quad L t \quad \frac{\sin N\gamma}{\sin \gamma} = \frac{L t}{\gamma \rightarrow m\pi} \frac{N \cos N\gamma}{\cos \gamma} = \pm N$$

For I to be maximum $\sin \gamma = 0$

$$\Rightarrow \gamma = 0, \pm\pi, \pm2\pi, \pm3\pi, \pm4\pi, \pm5\pi, \dots, \pm m\pi$$

$$\gamma = \pm m\pi = \frac{\pi}{\lambda} (d \sin \theta) \Rightarrow \sin \theta = \frac{m\lambda}{d}$$

$$\boxed{\Rightarrow d \sin \theta = \pm m\lambda} \quad \text{Where, } m = 0, 1, 2, 3, \dots$$

$$\Rightarrow d \sin \theta = \pm m \lambda$$

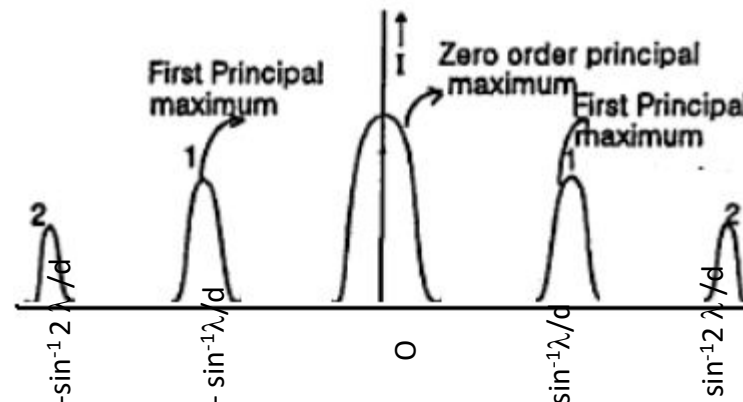
These maxima are known as **principal maxima** and 'm' is the order of principal maxima. Thus

$m = 0$ (at $\theta = 0$): **Zero order principal maximum.**

$m = 1$: **First order principal maxima.** $\theta_1 = \pm \sin^{-1} \frac{\lambda}{d}$

$m = 2$: **Second order principal maxima.** $\theta_2 = \pm \sin^{-1} \frac{2\lambda}{d}$

$m = 3$: **Third order principal maxima.** $\theta_3 = \pm \sin^{-1} \frac{3\lambda}{d}$



Intensity of the principal maxima

General equation of intensity is given as:

$$I = \frac{A^2 \sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma}$$

$$\Rightarrow \gamma = 0, \pm\pi, \pm2\pi, \pm3\pi, \pm4\pi, \pm5\pi, \dots, \pm m\pi$$

Substituting $\gamma = \pm m\pi$ we get
$$Lt_{\gamma \rightarrow m\pi} \frac{\sin N\gamma}{\sin \gamma} = Lt_{\gamma \rightarrow m\pi} \frac{N \cos N\gamma}{\cos \gamma} = \pm N$$

Thus intensity of principal maxima is given as

$$I = N^2 \frac{A^2 \sin^2 \beta}{\beta^2}$$

$$I = \frac{N^2 A^2 \sin^2 \beta}{\beta^2} \quad \text{If } \beta \rightarrow 0 \Rightarrow I = N^2 A^2$$

MINIMA

Recall, the intensity of diffraction pattern is given as

$$I = \frac{A^2 \sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma}$$

The intensity will be zero when either

$$\sin N\gamma = 0 \text{ but not } \sin \gamma = 0$$

$$\text{or} \quad \sin \beta = 0 \text{ but not } \beta = 0$$

$$\sin N\gamma = 0 \quad \Rightarrow \quad N\gamma = 0, \pm\pi, \pm2\pi, \pm3\pi, \dots = \pm n\pi$$

$$\Rightarrow N \frac{\pi}{\lambda} d \sin \theta = \pm n\pi \quad \Rightarrow \quad Nd \sin \theta = \pm n\lambda$$

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Where n can have the all integer values except 0, N, 2N, 3N.....

For example, if $n = N$ then,

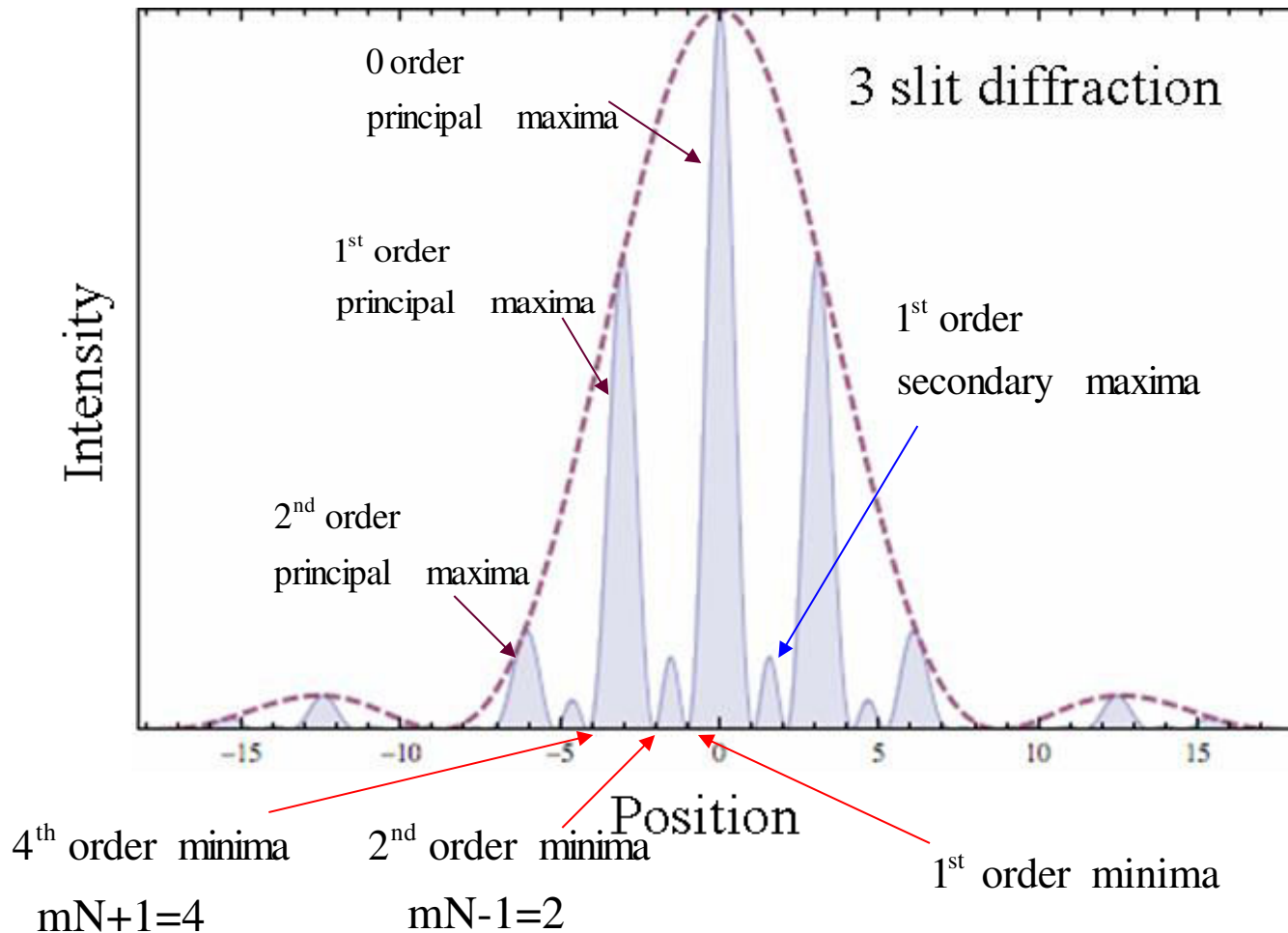
$$\cancel{Nd} \sin \theta = \pm \cancel{N}\lambda \quad \Rightarrow d \sin \theta = \pm \lambda$$

Which is condition for 1st order principal maxima. Similar is the case with $n = 0, 2N, 3N, \dots$

From the above equation it is clear that

- (i) $n = 0$ gives the principal maxima of zero order;
- (ii) $n = 1, 2, 3, \dots, (N-1)$ gives minima;
- (iii) $n = N$ gives principal maxima of the first order.

Thus we may say that there are $N-1$ equally spaced minima between zero order and first order principal maxima or between two consecutive principal maxima. Further, between two minima there would be maxima known as **secondary maxima**. The number of secondary maxima would be $N-2$.



$$N = 3$$

Example: For $a=2b$, 5 consecutively placed slits are considered for diffraction i.e. $N=5$. Find out the minima b/w two principal maxima, secondary maxima and principal maxima in central maxima

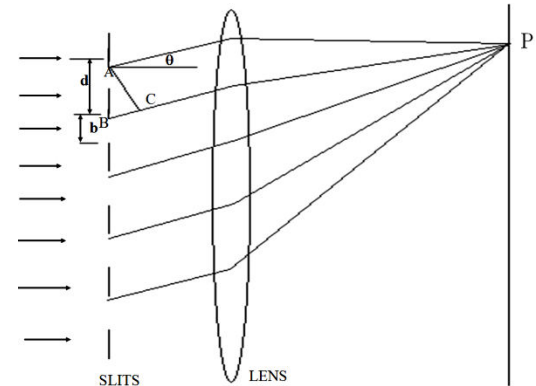
Principal maxima $d \sin \theta = \pm m \lambda$ $m = 0, 1, 2..$

Minima $5d \sin \theta = \pm n \lambda$ $n = 0, 1, 2, 3, 4, 5$

No. of minima between two consecutive principal maxima = 4

No. of secondary maxima = 3

No. of principal maxima in the central maxima = 5 (-2,-1,0,1,2)



SECONDARY MAXIMA

Intensity is given as
$$I = \frac{A^2 \sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma}$$

To find out position of secondary maxima, let us again differentiate

$$\text{i.e. } \frac{d}{d\gamma} \left[\left(\frac{\sin N\gamma}{\sin \gamma} \right)^2 \right] = \frac{2 \sin N\gamma}{\sin \gamma} \left[\frac{N \sin \gamma \cos N\gamma - \sin N\gamma \cos \gamma}{\sin^2 \gamma} \right] = 0$$

$$\Rightarrow \frac{\sin N\gamma}{\sin \gamma} = 0 \quad \text{or} \quad N \sin \gamma \cos N\gamma - \sin N\gamma \cos \gamma = 0$$

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$$\Rightarrow N \sin \gamma \cos N\gamma = \sin N\gamma \cos \gamma$$

$$\Rightarrow N \tan \gamma = \tan N\gamma$$

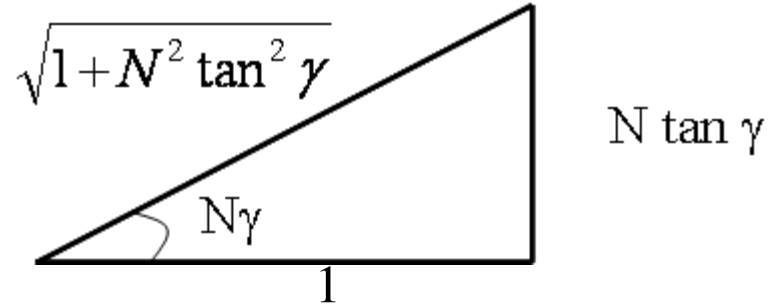
Again using graphical method one can derive condition for secondary maxima.

Intensity of secondary diffraction maxima from N slits

Given that $\tan N\gamma = N \tan \gamma = \frac{N \tan \gamma}{1}$

From the triangle

$$\sin N\gamma = \frac{N \tan \gamma}{\sqrt{1 + N^2 \tan^2 \gamma}}$$



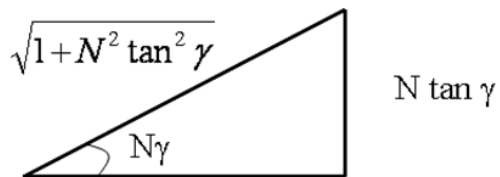
$$\Rightarrow \frac{\sin N\gamma}{\sin \gamma} = \frac{N \cancel{\sin \gamma}}{\cancel{\sin \gamma} \cos \gamma \sqrt{1 + \frac{N^2 \sin^2 \gamma}{\cos^2 \gamma}}} = \frac{N}{\frac{\cancel{\cos \gamma}}{\cos \gamma} \sqrt{\cos^2 \gamma + N^2 \sin^2 \gamma}}$$

Intensity of secondary diffraction maxima from N slits

Given that $\tan N\gamma = N \tan \gamma = \frac{N \tan \gamma}{1}$

From the triangle

$$\sin N\gamma = \frac{N \tan \gamma}{\sqrt{1 + N^2 \tan^2 \gamma}}$$



$$\Rightarrow \frac{\sin N\gamma}{\sin \gamma} = \frac{N}{\sqrt{\cos^2 \gamma + N^2 \sin^2 \gamma}}$$

$$\Rightarrow \frac{\sin N\gamma}{\sin \gamma} = \frac{N \sin \gamma}{\cancel{\sin \gamma} \cos \gamma \sqrt{1 + \frac{N^2 \sin^2 \gamma}{\cos^2 \gamma}}} = \frac{N}{\cancel{\cos \gamma} \sqrt{\cos^2 \gamma + N^2 \sin^2 \gamma}}$$

$$\frac{\sin N\gamma}{\sin \gamma} = \frac{N}{\sqrt{1 - \sin^2 \gamma + N^2 \sin^2 \gamma}} = \frac{N}{\sqrt{(N^2 - 1) \sin^2 \gamma + 1}}$$

Or, $\frac{\sin^2 N\gamma}{\sin^2 \gamma} = \frac{N^2}{(N^2 - 1) \sin^2 \gamma + 1}$

Therefore the intensity of secondary maxima will be

$$I' = \frac{A^2 \sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma} = \frac{A^2 N^2 \sin^2 \beta}{\beta^2 [(N^2 - 1) \sin^2 \gamma + 1]}$$

But intensity of principal maxima is given as $I = \frac{A^2 N^2 \sin^2 \beta}{\beta^2}$

Therefore,

$$\frac{I'}{I} = \frac{\frac{A^2 N^2 \sin^2 \beta}{\beta^2 [(N^2 - 1) \sin^2 \gamma + 1]}}{\frac{A^2 N^2 \sin^2 \beta}{\beta^2}}$$

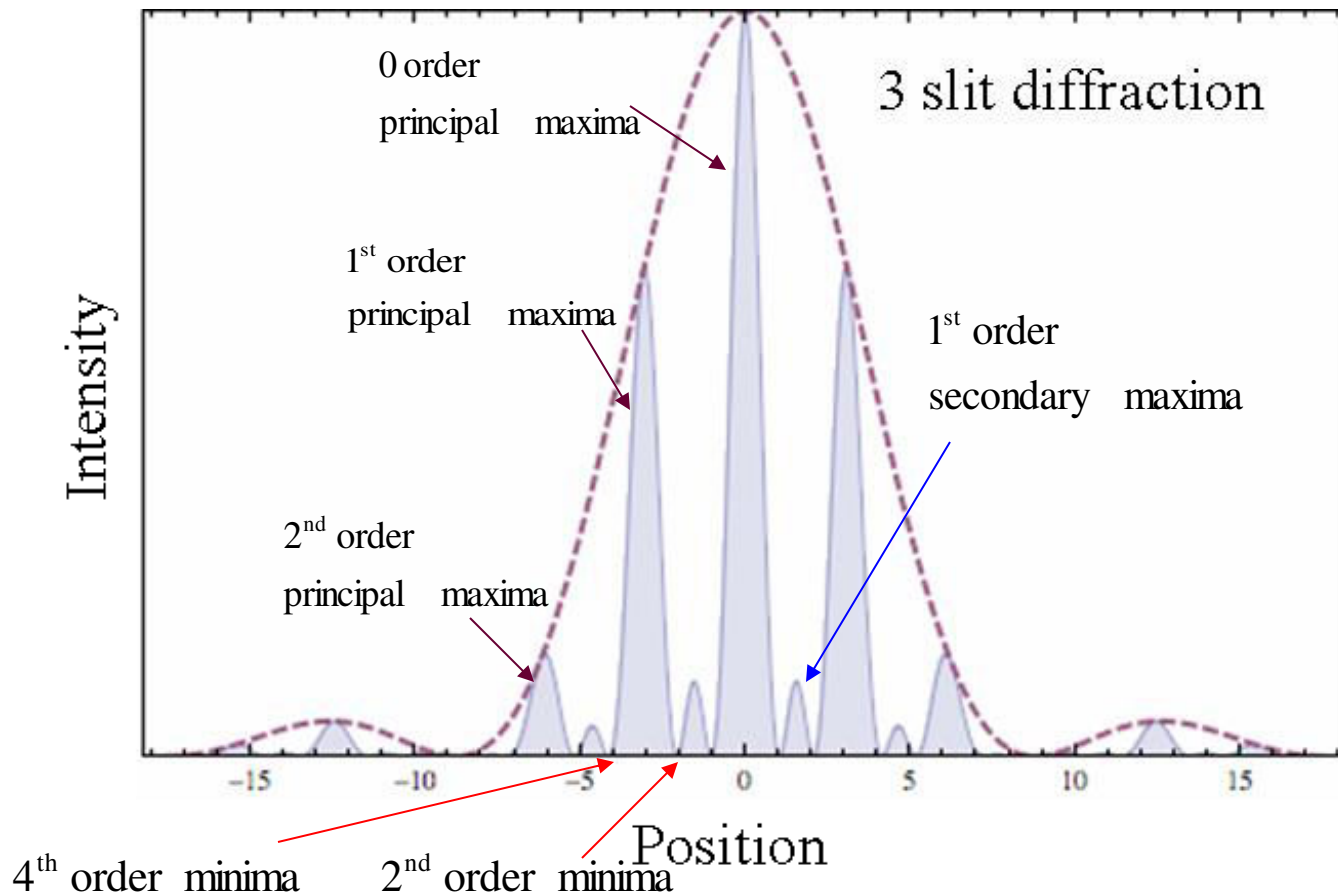
$$\boxed{\frac{I'}{I} = \frac{1}{(N^2 - 1) \sin^2 \gamma + 1}}$$

Conclusion: Larger the number of slits, smaller the intensity of secondary maxima.

Example: A diffraction grating, (normally 1500 lines/ inch), secondary maxima are not visible in the diffraction pattern.

WIDTH OF A PRINCIPAL MAXIMA

Angular width of a principal maxima is equal to the angular separation between first two minima on either side of the principal maxima.

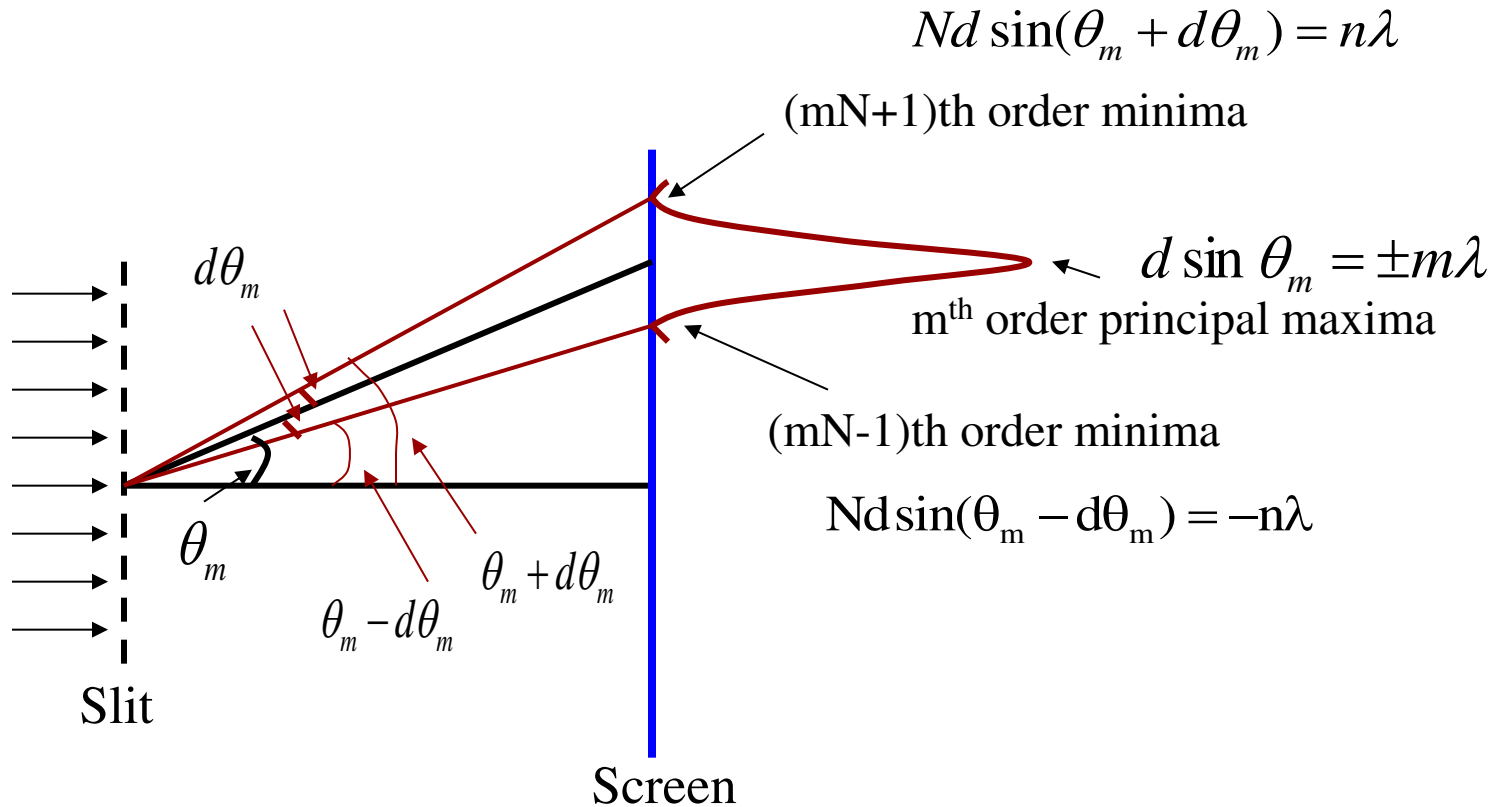


$$m = 1$$

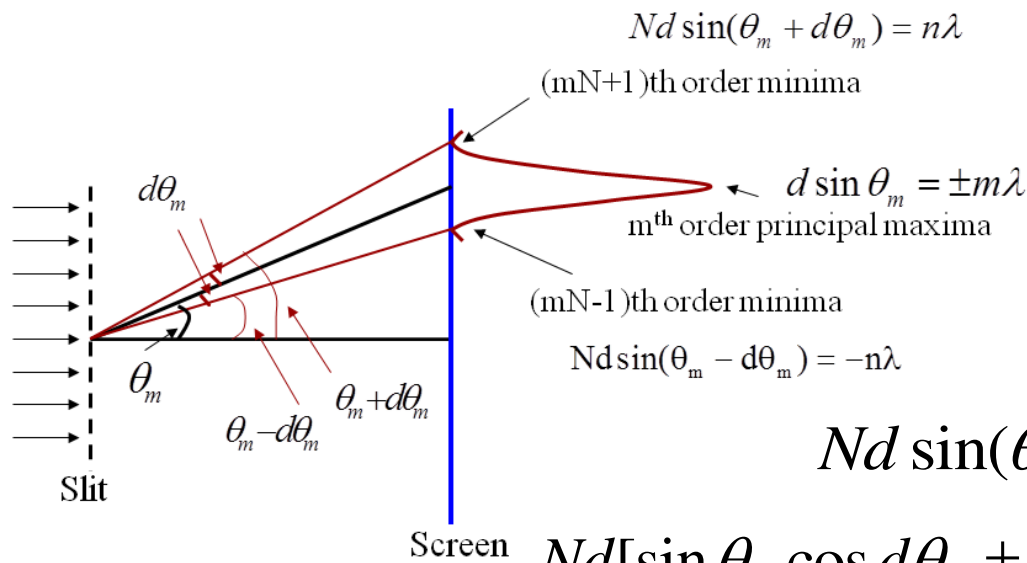
$$N = 3$$

$$mN \pm 1 = 4 \text{ and } 2$$

WIDTH OF A PRINCIPAL MAXIMA



WIDTH OF A PRINCIPAL MAXIMA



$$Nd \sin(\theta_m + d\theta_m) = n\lambda$$

$(mN+1)^{\text{th}}$ order minima

$$d \sin \theta_m = \pm m\lambda$$

m^{th} order principal maxima

$(mN-1)^{\text{th}}$ order minima

$$Nd \sin(\theta_m - d\theta_m) = -n\lambda$$

$$Nd \sin(\theta_m \pm d\theta_m) = \pm n\lambda$$

$$Nd \sin(\theta_m \pm d\theta_m) = \pm(mN \pm 1)\lambda$$

$$Nd[\sin \theta_m \cos d\theta_m \pm \cos \theta_m \sin d\theta_m] = \pm(mN \pm 1)\lambda$$

For very small $d\theta_m$

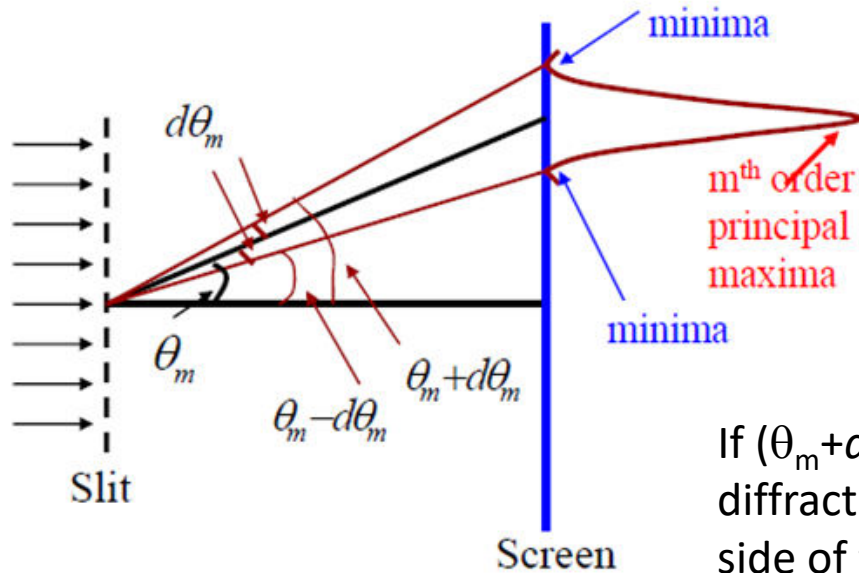
$$\cos d\theta_m \approx 1 \quad \text{and} \quad \sin d\theta_m \approx d\theta_m$$

Thus,
$$Nd[\sin \theta_m \pm \cos \theta_m d\theta_m] = \pm mN\lambda \pm \lambda$$

But,
$$d \sin \theta_m = \pm m\lambda$$

Therefore,
$$\pm mN\lambda \pm Nd \cos \theta_m d\theta_m = \pm mN\lambda \pm \lambda$$

Width of a Principal Maxima



For the diffraction through N slits, m^{th} order principal maximum occurs at

$$d \sin \theta_m = \pm m \lambda \quad \text{where } m = 0, 1, 2, 3, \dots \quad \text{.....(1)}$$

Further, the minima occur at the angles given by

$$d \sin \theta = \pm n \lambda / N$$

If $(\theta_m + d\theta_m)$ and $(\theta_m - d\theta_m)$ represent the angle of diffraction corresponding to the first minimum on either side of the principal maximum, then

$$Nd \sin(\theta_m + d\theta_m) = \pm n \lambda \Rightarrow Nd \sin(\theta_m + d\theta_m) = \pm(mN \pm 1) \lambda \quad \text{.....(2)}$$

For very small $d\theta_m$ $\cos d\theta_m \approx 1$ & $\sin d\theta_m \approx d\theta_m$

On simplifying, we will get

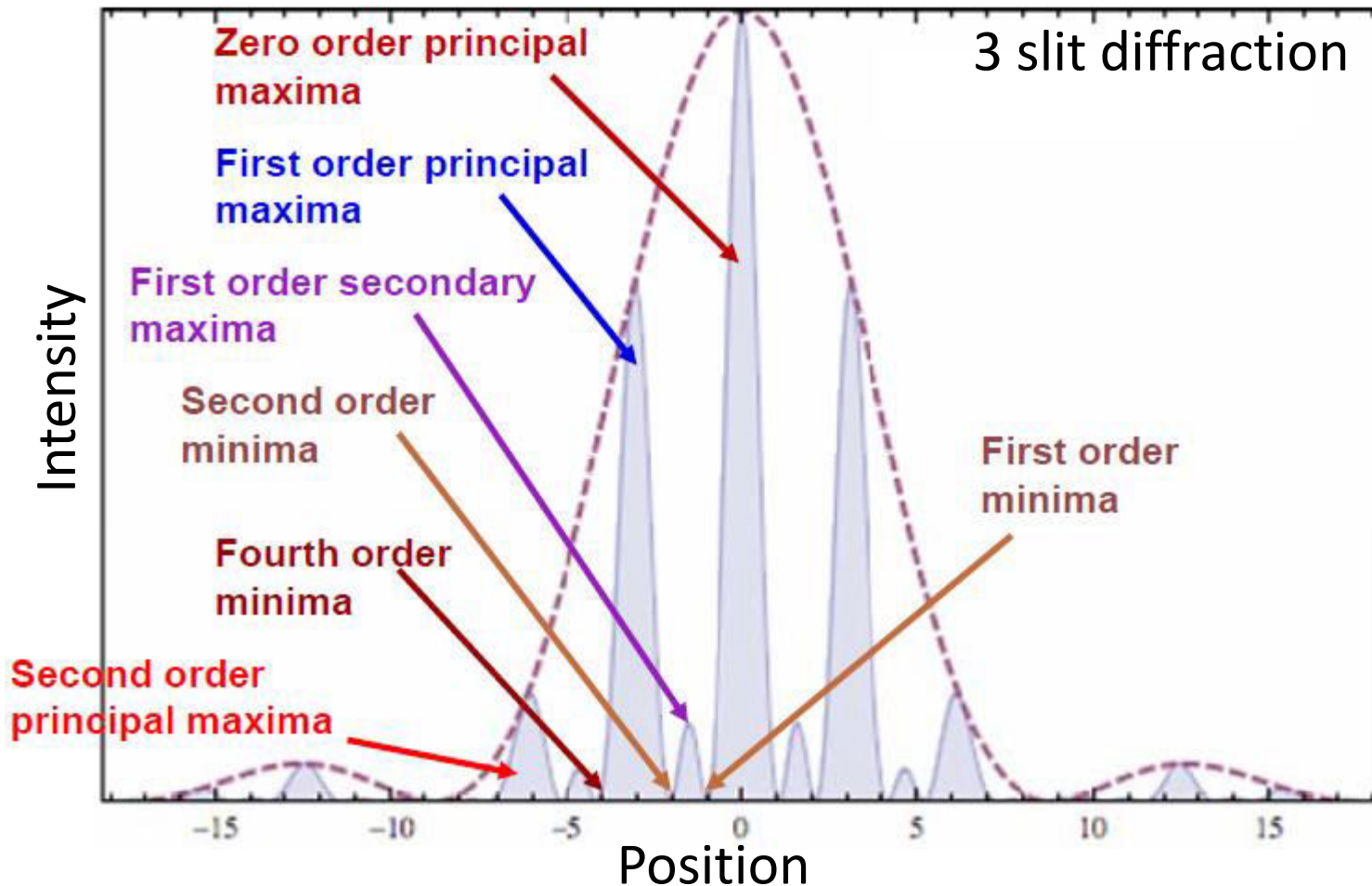
$$2d\theta_m = \frac{2\lambda}{Nd \cos \theta_m}$$

Here, $2d\theta_m$ is the width of m^{th} order principal maxima.

it is clear that **larger the number slits sharper will be principal maxima.**

Width of a Principal Maxima

Angular width of a principal maxima is equal to the angular separation between first two minima on either side of the principal maxima.



$m=1$
1st principal
maxima

$N=3$
no. of slits

2nd order minima
 $(mN-1) = 2$

4th order minima
 $(mN+1) = 4$

Overlapping Orders in Diffraction Grating

If $\lambda_1 = 4000 \text{ \AA}$ and $\lambda_2 = 5000 \text{ \AA}$

$$d \sin \theta = \pm m \lambda$$

$$\Rightarrow \sin \theta_1 = \pm \frac{4000 m_1}{d} \text{ and}$$

$$\sin \theta_2 = \pm \frac{5000 m_2}{d}$$

If $m_1 = 5$ and $m_2 = 4$, then $\theta_1 = \theta_2$.

5th order of principal maxima of $4000 \text{ \AA}$ will coincide with 4th order principal maxima of $5000 \text{ \AA}$ wavelength.

Highest Possible Order of Principal Maximum

$$d \sin \theta = m\lambda$$

maximum value of $\theta = \pi/2$, which will give

$$m = d/\lambda$$

Here m is the highest order for principal maximum visible. If m is not an integer, then the highest possible order is given by the integer lower than m .

Thus, the grating element $d = (b + a)$ determines the maximum possible order.

❑ In the equation $d \sin \theta = m\lambda$ if $d < \lambda$ then $\sin \theta > 1$ **NOT POSSIBLE**
i.e. first order spectrum will be absent.

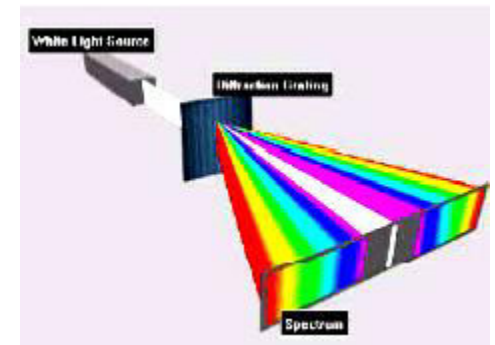
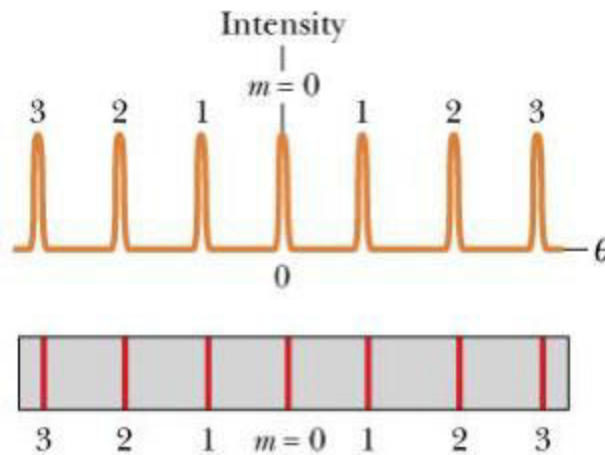
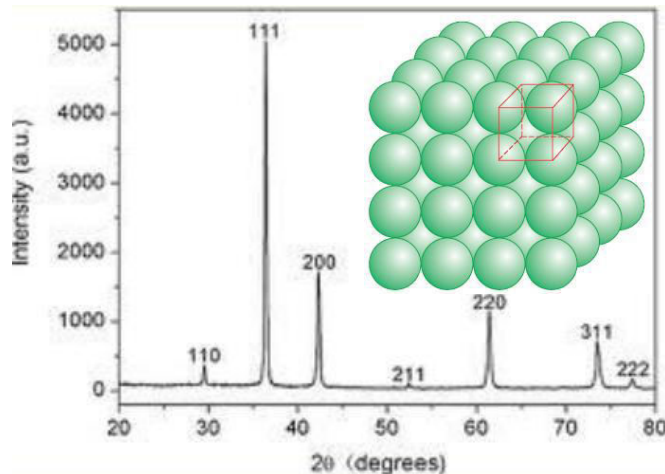
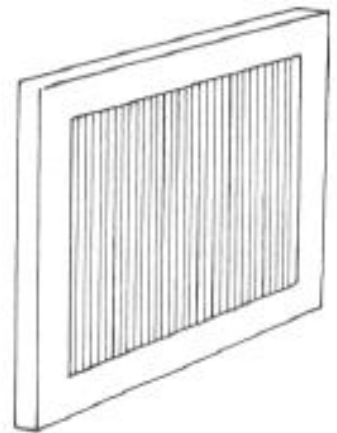
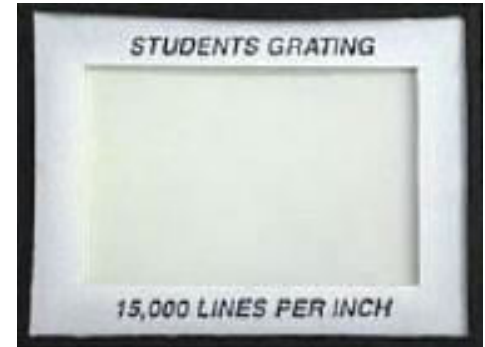
❑ Similarly second, third etc. orders will be absent if $d < 2\lambda$, $d < 3\lambda$, respectively.
In general if $d < m\lambda$ m^{th} order spectrum will be absent.

Diffraction Grating

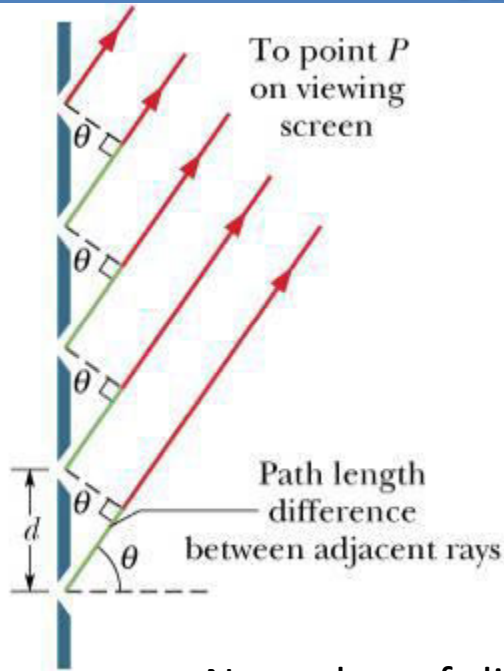
Like the double slit arrangement but with a much greater number of slits, or rulings, Sometimes as many as several 1000 per millimeter.

Light passed through the grating forms narrow interference fringes that can be analyzed to determine the wavelength

As the number of rulings increases beyond 2 the intensity plot changes from that of a double slit pattern to one with very narrow maxima (called lines) surrounded by **relatively wide dark regions**.



Diffraction Grating (N-slits)



Condition for principal maxima:

$$d \sin \theta_m = m\lambda$$

where $m = 0, 1, 2, 3, \dots$

θ_m represents position of line and the integers m are called order numbers.

N number of slits = numbers of lines per inch or mm.

Typically, there are 15000 lines in 2.54 cm in the laboratory grating.

Thus only two orders are possible in grating for 6000Å wavelength.

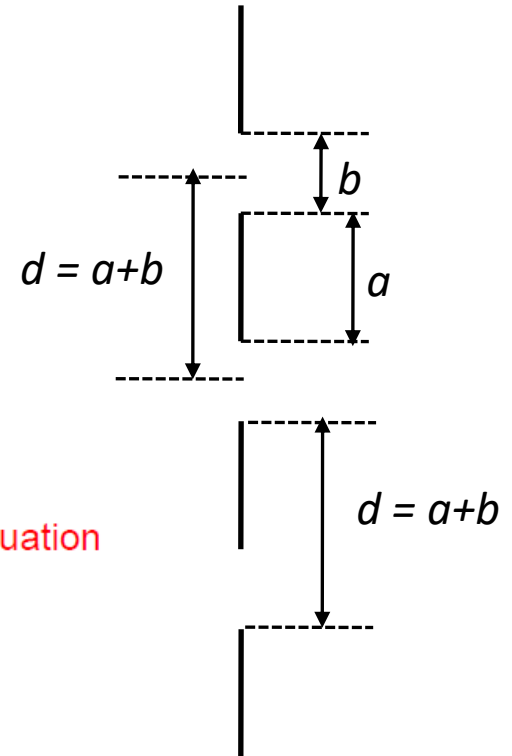
From grating equation, by knowing the angle of diffraction, one can determine wavelength of different colors.

$$d = \frac{2.54}{15000} = 1.693 \times 10^{-4} \text{ cm}$$

$$d \sin \theta_m = \pm m\lambda \text{ - Grating equation}$$

$$\Rightarrow m = \frac{d \sin \theta_m}{\lambda}$$

$$= \frac{1.693 \times 10^{-4}}{6 \times 10^{-5}} = 2.8$$



Dispersive Power of a Grating

$$\lambda \rightarrow \lambda + d\lambda, \theta \rightarrow \theta + d\theta$$

Rate of change of angle of diffraction with respect to change of wavelength $d\theta/d\lambda$

Grating Equation: $d \sin \theta_m = \pm m \lambda$
on differentiating $d \cos \theta_m d\theta_m = \pm m d\lambda \Rightarrow \boxed{\frac{d\theta_m}{d\lambda} = \frac{\pm m}{d \cos \theta_m} = \frac{1}{\lambda \cot \theta_m}}$

The angular dispersion defined as the angular separation per unit wavelength.

If θ is very small
$$d\theta_m = \pm \frac{m}{d} d\lambda$$

For small values of θ and a given value of m , difference in angle of diffraction is directly proportional to the difference in wavelength. (i.e. $\Delta\theta/\Delta\lambda$ is a constant, such a spectrum is known as **normal spectrum**)

For large θ , dispersion is greater at red end of the spectrum.

Dispersive power:

- higher for higher order maxima
- higher for small grating element or large N
- more for small wave length in the same order maxima

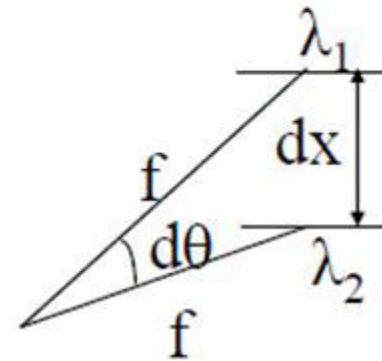
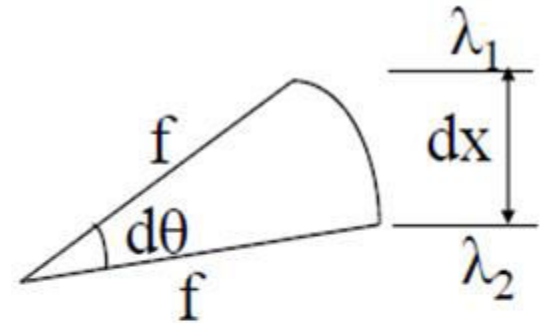
Linear Dispersive Power

If dx is the linear separation of two spectral lines differing in wavelength $d\lambda$ in the focal plane of a lens of focal length f , then

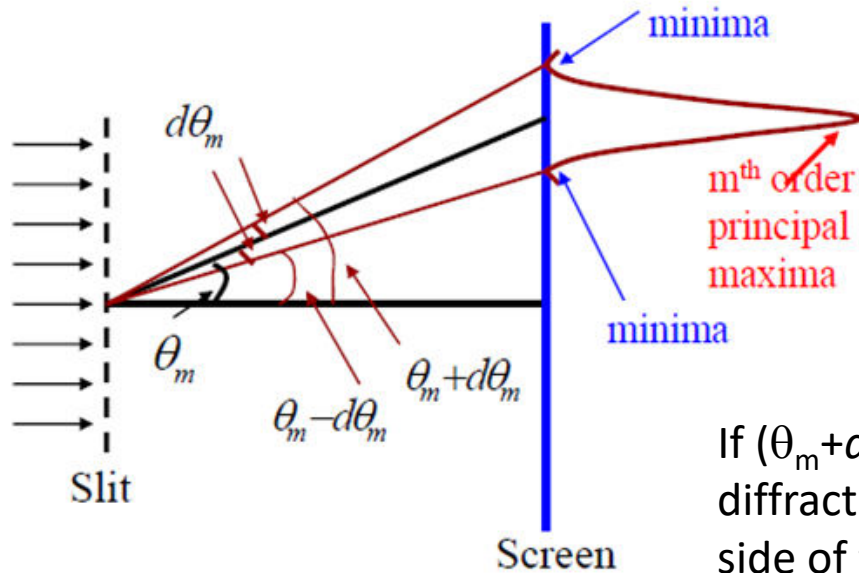
$$dx = fd\theta$$

$$\text{but } \frac{d\theta}{d\lambda} = \pm \frac{m}{d \cos \theta} \quad \text{from angular dispersion}$$

$$\Rightarrow \frac{dx}{d\lambda} = \pm f \frac{m}{d \cos \theta}$$



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Further, the minima occur at the angles given by

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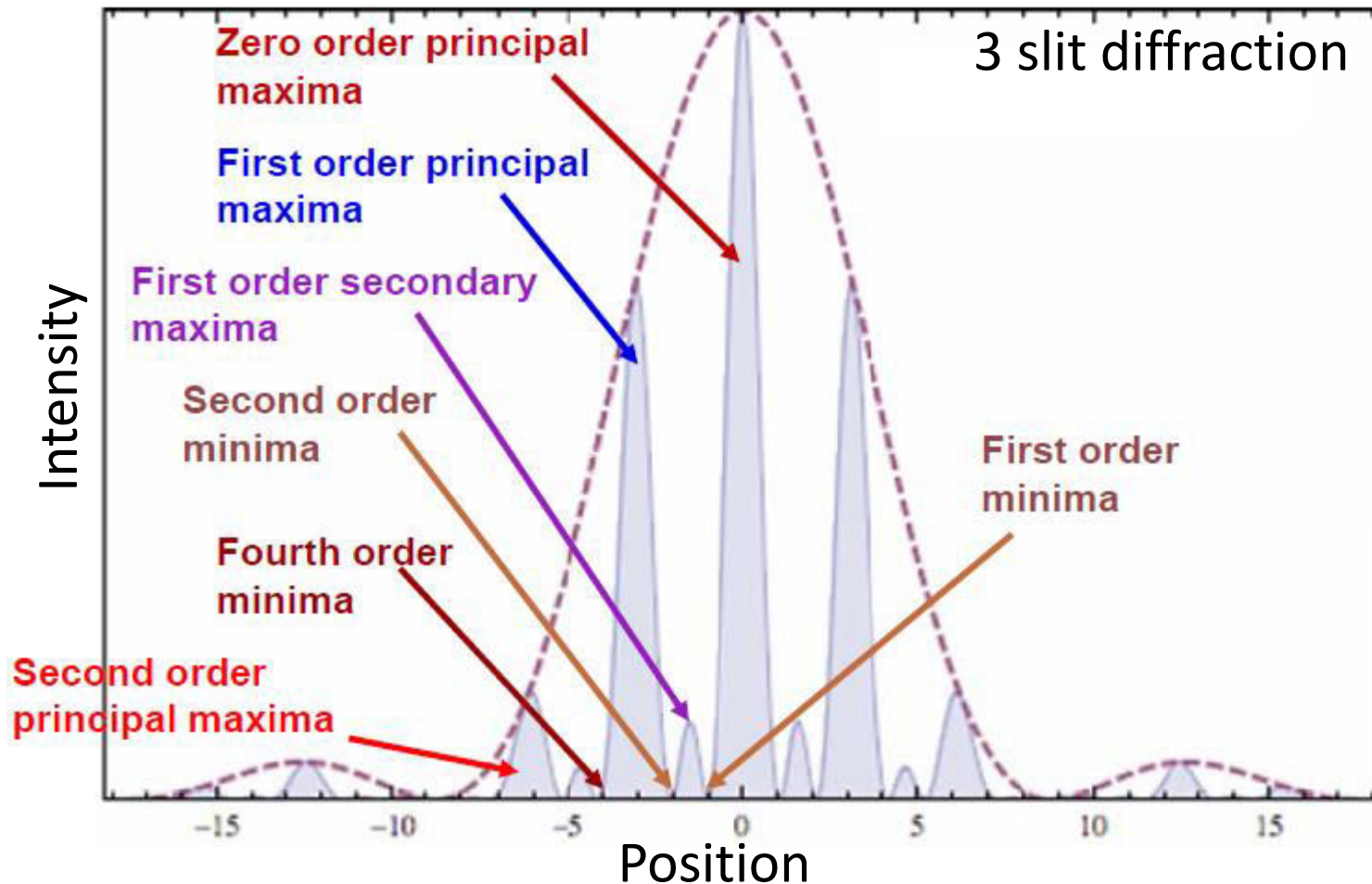
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❑ In the equation $d \sin \theta = m\lambda$ if $d < \lambda$ then $\sin \theta > 1$ **NOT POSSIBLE**
i.e. first order spectrum will be absent.

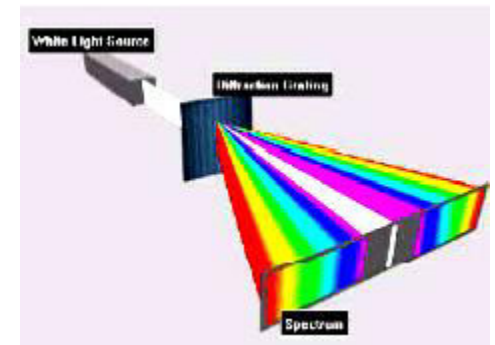
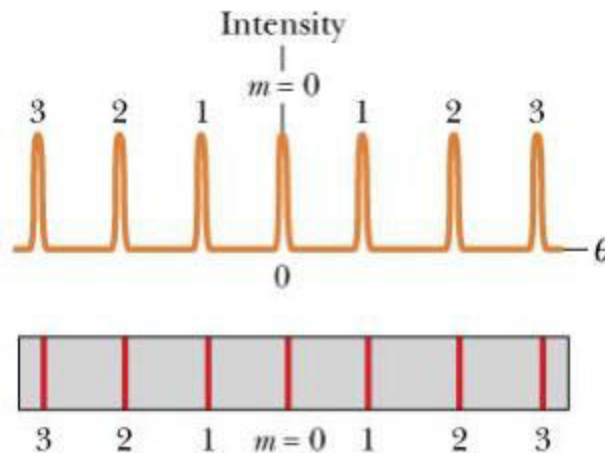
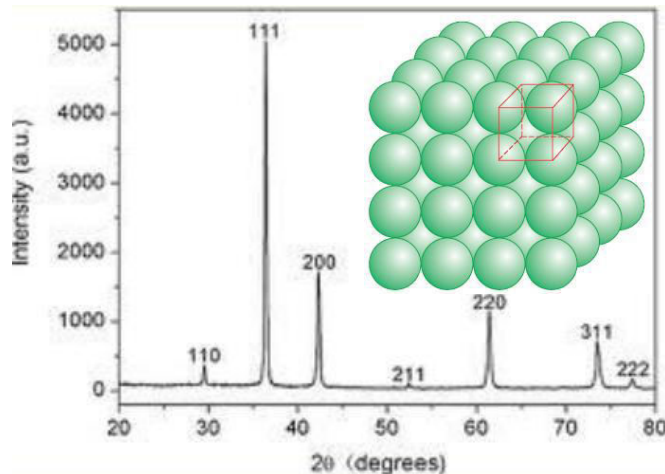
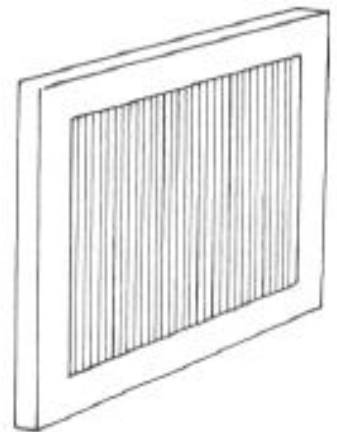
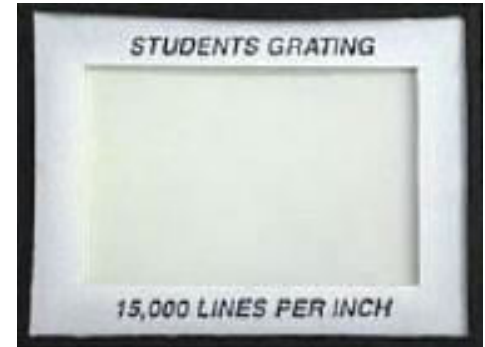
❑ Similarly second, third etc. orders will be absent if $d < 2\lambda$, $d < 3\lambda$, respectively.
In general if $d < m\lambda$ m^{th} order spectrum will be absent.

Diffraction Grating

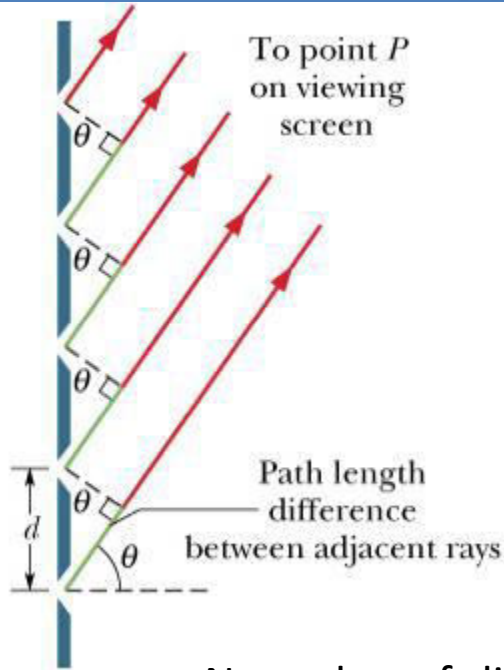
Like the double slit arrangement but with a much greater number of slits, or rulings, Sometimes as many as several 1000 per millimeter.

Light passed through the grating forms narrow interference fringes that can be analyzed to determine the wavelength

As the number of rulings increases beyond 2 the intensity plot changes from that of a double slit pattern to one with very narrow maxima (called lines) surrounded by **relatively wide dark regions**.



Diffraction Grating (N-slits)



Condition for principal maxima:

$$d \sin \theta_m = m\lambda$$

where $m = 0, 1, 2, 3, \dots$

θ_m represents position of line and the integers m are called order numbers.

N number of slits = numbers of lines per inch or mm.

Typically, there are 15000 lines in 2.54 cm in the laboratory grating.

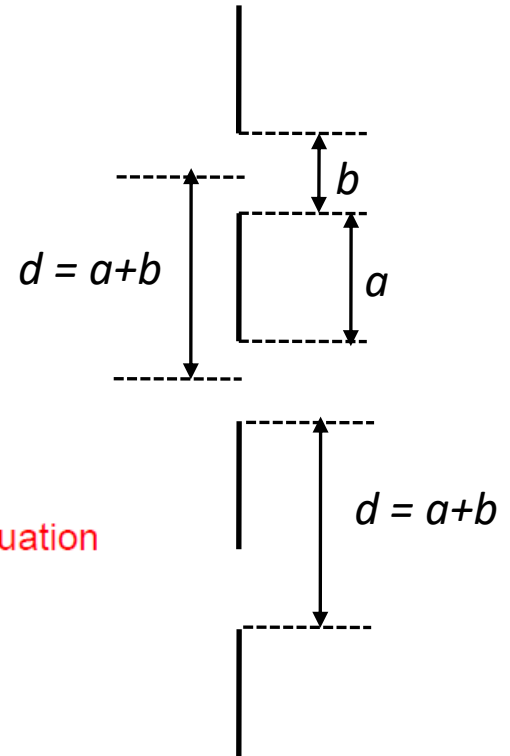
Thus only two orders are possible in grating for 6000Å wavelength.

From grating equation, by knowing the angle of diffraction, one can determine wavelength of different colors.

$$d = \frac{2.54}{15000} = 1.693 \times 10^{-4} \text{ cm}$$

$$d \sin \theta_m = \pm m\lambda \text{ - Grating equation}$$

$$\begin{aligned} \Rightarrow m &= \frac{d \sin \theta_m}{\lambda} \\ &= \frac{1.693 \times 10^{-4}}{6 \times 10^{-5}} = 2.8 \end{aligned}$$



Dispersive Power of a Grating

$$\lambda \rightarrow \lambda + d\lambda, \theta \rightarrow \theta + d\theta$$

Rate of change of angle of diffraction with respect to change of wavelength $d\theta/d\lambda$

Grating Equation: $d \sin \theta_m = \pm m \lambda$
on differentiating $d \cos \theta_m d\theta_m = \pm m d\lambda \Rightarrow$

$$\frac{d\theta_m}{d\lambda} = \frac{\pm m}{d \cos \theta_m} = \frac{1}{\lambda \cot \theta_m}$$

The angular dispersion defined as the angular separation per unit wavelength.

If θ is very small
$$d\theta_m = \pm \frac{m}{d} d\lambda$$

For small values of θ and a given value of m , difference in angle of diffraction is directly proportional to the difference in wavelength. (i.e. $\Delta\theta/\Delta\lambda$ is a constant, such a spectrum is known as **normal spectrum**)

For large θ , dispersion is greater at red end of the spectrum.

Dispersive power:

- higher for higher order maxima
- higher for small grating element or large N
- more for small wave length in the same order maxima

Linear Dispersive Power

If dx is the linear separation of two spectral lines differing in wavelength $d\lambda$ in the focal plane of a lens of focal length f , then

$$dx = fd\theta$$

$$\text{but } \frac{d\theta}{d\lambda} = \pm \frac{m}{d \cos \theta} \quad \text{from angular dispersion}$$

$$\Rightarrow \frac{dx}{d\lambda} = \pm f \frac{m}{d \cos \theta}$$

